

Is it possible to verify this theory experimentally? We are thinking, and trying, but there is as yet no experimental verification. There are, however, a number of different attempts.

One of these derives from the study of black holes. In the heavens we can now observe black holes formed by collapsed stars. Crushed by its own weight the matter of these stars has collapsed upon itself and disappeared from our view. But where has it gone? If the theory of loop quantum gravity is correct, matter cannot really have collapsed to an infinitesimal point, because infinitesimal points do not exist – only finite chunks of space. Collapsing under its own weight, matter must have become increasingly dense, up to the point where quantum mechanics must have exerted a contrary, counterbalancing pressure.

This hypothetical final stage in the life of a star, where the quantum fluctuations of space-time balance the weight of matter, is what is known as a ‘Planck star’. If the sun were to stop burning and to form a black hole it would measure about one and a half kilometres in diameter. Inside this black hole the sun’s matter would continue to collapse, eventually becoming such a Planck star. Its dimensions would then be similar to those of an atom. The